N-gram Hints in Language Model Training

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Abstract

An n-gram hint tells a language model which token most often followed the last n tokens in the training data. Hints recently cut the NanoGPT Speedrun record from 39.9 s to 6.7 s, but training uses only hints from tokens trained in earlier steps, because a large corpus from the start makes the model rely on hints too much. In small Muon-trained models we show that this failure comes from a token's own occurrence in the counts: keeping it gives a validation loss of 3.53, while removing it and dropping half of the hints gives 1.80, against 1.89 for hints from earlier steps. On web text the token's whole page must be removed. That fix repairs validation loss but not learning. With its own occurrence removed, a model that always gets hints still memorizes almost nothing itself, and hints from a full table at all three places stop it from learning comparison. In a small test, a model that learned comparison first kept more of it after hints were added than a model trained with hints from the start. Training with 5% to 60% of the table and testing with all of it kept comparison at .97 or above and raised facts from .81 to about .88, with no switch and no second pass, which supports the record's choice of a small corpus in training. With test counts scaled, the limit follows how many copies of each training sentence the table holds, about 5, not table size. When one model trained on web text and reasoning together, a switch trained to predict whether each hint is right raised comparison from chance to at least .905 at three model sizes and lowered web text loss below plain hints by 0.04 to 0.09 at the longest training we ran. On a finished 2.5-billion-parameter model, whose weights we left unchanged, hints added 1.4 points to next-word accuracy on unseen Wikipedia text, recalled 6 of 11 new facts in their stored wording and 1 of 11 when reworded, and left comparison answers unchanged. On code from one source they added 1.7 to 17.7 points when the model read 128 tokens, and 0.8 to 8.6 points in the two cases rerun at 512 tokens; the gain rose with how often exact 6-token phrases repeat and with how low the model scored alone. All other models are small and each setting ran once. The code is public [23].

1 Introduction

Language models learn slowly from text they have seen only a few times, and web text is full of passages that repeat only occasionally. Brunborg [1] uses this: during training, idle CPUs build a corpus that records which tokens follow each n-gram. The model receives the most common next token as a hint and learns how much to trust it. The record went from 39.9 s to 6.7 s [2]. The blog post reports two things that matter for this paper. A large hint corpus from the start makes the backbone rely on the hints too much, so training uses a small corpus and validation a large one. And each doubling of the corpus cuts the loss more than the last [1].

The record's code computes training hints in a strictly causal way: a step's hints count only tokens trained in earlier steps [3]. That has a cost. At validation the model meets hints from a far larger corpus than any it trained with, so it must trust counts it has never seen.

We ask whether a language model trained with Muon can use hints from the whole corpus from the first step, without the failure the blog post describes, and what hints do to memorizing and reasoning. We make four contributions.

1. We show that a token's own occurrence in the counts is enough to break training with a full corpus.
2. We show that removing it, plus random hint dropout, beats causal hints at three model sizes on Shakespeare.
3. We measure how the gain depends on the ratio of corpus size to tokens trained.
4. We test what hints help, facts or reasoning, find what makes hints block reasoning, and test fixes on web text.
5. We measure what hints do for a finished model whose weights stay unchanged.

The main question is when a lookup table stops a model from learning. The answer has two parts. Removing a token's own occurrence fixes the failure in validation loss, so by that measure a full table is safe. It does not protect the model's own learning: with the occurrence removed, a full table in training still stops the model from memorizing and from learning comparison, and a small table in training, the record's choice, keeps comparison. So leave-one-out answers the loss problem, and Section 8 answers the learning problem.

2 Background

Muon. Muon [4] is an optimizer for the matrices of a network. It keeps momentum, replaces the update with its nearest orthogonal matrix using Newton-Schulz steps, and moves the weights by a fixed step [5]. Wang et al. [6] find that Muon learns rare classes better than Adam, the same rare-data problem hints address. Embeddings stay on Adam.

Memory by lookup. Engram [7] adds hashed n-gram embedding tables whose vectors are learned, and Tensorizing Engram [8] and Lngram v2 [9] extend this line. Hints differ: nothing is learned per n-gram, and the corpus stores only the next tokens. RETRO [10] retrieves whole text chunks from a very large database.

Leakage. A count feature that includes the row's own label leaks the label, as in target encoding [11]. CatBoost [12] fixes this with ordered statistics, which use only earlier rows, the same idea as causal hints. Leave-one-out statistics are the other common fix, and RETRO filters out neighbours from the same document [10]. Our searches found no paper comparing causal and leave-one-out hints for n-gram hints, and none using hint dropout, which follows dropout [13].

3 Method

Model. We use the Muon lab model [14]: a character-level transformer on Tiny Shakespeare with RoPE, QK-norm, RMSNorm, a SwiGLU MLP, no biases and tied embeddings, with 4 layers and 4 heads. The first 90% of the text is the training split. Muon trains the hidden matrices with momentum 0.95, and Adam trains the embeddings and norm weights at 0.003. Training has 20 warmup steps and a linear cooldown over the last 30%.

Hints. We follow the blog post, with lengths 6, 12 and 24 characters in place of 6, 10 and 20 tokens. For each length the hint is the most common next character. Its count falls into one of 7 buckets and its purity (the share of the n-gram's occurrences followed by that character) into one of 6, giving 126 kinds. Each kind has a learned amount σ and a learned vector β. The hint vector is

$$H_t = \sum_n \big(\sigma_{k(n,t)}\, E[\text{hint}_n(t)] + \beta_{k(n,t)}\big), \tag{1}$$

where E is the embedding table. We add H_t after the embedding, after layer 2 and before the output head, each with its own learned scale. The hint parameters use Adam at 0.03.

Training hint policies. Validation always uses the whole training split as its corpus. Training uses one of these:

- **None:** no hints.
- **Own token kept:** the whole training split, with each token's own occurrence still in the counts.
- **Earlier steps:** only tokens trained in earlier steps, as in the record.
- **Leave-one-out:** the whole training split with each token's own occurrence removed. If removing it leaves no count, there is no hint, exactly as for an unseen n-gram at validation.
- **Leave-one-out with dropout:** the same, and each hint is dropped with probability 0.5.

Training streams shuffled, non-overlapping chunks of 65 characters, so a token appears at most once per pass. The default batch is 8 and the default length is 200 steps, or 102K trained tokens. The corpus is 1.0M characters, 10 times the tokens trained. The record's ratio is about 520 times [1], which a 1 MB text cannot reach.

Measurement. Each setting was run once at Muon learning rates 0.01, 0.02 and 0.04, and we report the best, on 96 fixed validation chunks.

4 Results

4.1 Keeping the token's own occurrence breaks training

Table 1 shows the results at width 64. With the token's own occurrence in the counts, every training n-gram has a match, and a long n-gram usually has only that one occurrence, so the hint is the answer. The model learns to trust hints it will not get at validation, and its loss with hints is 3.53 against 2.16 with no hints at all. Removing that one occurrence brings the loss to 1.84.

4.2 Hint dropout reduces the loss penalty with hints off

Leave-one-out alone makes the backbone worse when hints are off: 2.508 against 2.165 with no hints, which matches the blog post's report of relying on hints too much. Dropping half of the hints brings this to 2.256 and lowers the loss with hints from 1.843 to 1.805. Dropout also helps earlier-steps hints (1.890 to 1.860), but they stay behind. Dropping a quarter, half or three quarters gives losses within 0.007 of each other, so half is a default and not a tuned best. This measures overall loss only. On memorizing, dropout does far less: Table 8 shows that a model needs hints removed from 90% of its tokens to match a model that never saw hints.

Training hints	Validation, hints on	Validation, hints off
None	–	2.165
Own token kept	3.525	4.735
Earlier steps	1.890	2.307
Earlier steps, half dropped	1.860	2.204
Leave-one-out	1.843	2.508
Leave-one-out, quarter dropped	1.800	2.349
Leave-one-out, half dropped	**1.805**	**2.256**
Leave-one-out, three quarters dropped	1.807	2.206

Table 1: Width 64, 200 steps, batch 8. Best of three learning rates by loss with hints.

4.3 The gain stays the same as the model grows

Table 2 shows a gap between 0.070 and 0.085 across a 15-fold range of parameters.

Width	Parameters	No hints	Earlier steps	Leave-one-out, half dropped	Gap
32	56K	2.317	1.938	1.858	0.081
64	209K	2.165	1.890	1.805	0.085
128	812K	2.085	1.840	1.769	0.070

Table 2: Validation loss with hints on, 200 steps, batch 8, corpus 10 times the tokens trained.

4.4 The gain depends on corpus size divided by tokens trained

Table 3 changes the tokens trained at width 32 with the corpus fixed. The gap is 0.370 at a ratio of 40, 0.081 at 10 and 0 at 2.5, because hints from earlier steps approach the whole corpus as training goes on. At 3200 steps a second pass starts, so hints from earlier steps contain each token's own earlier occurrence and the failure above returns: loss rises to 2.533, while leave-one-out reaches 1.603. Causal hints are safe only within one pass.

Steps	Batch	Ratio	No hints	Earlier steps	Leave-one-out, half dropped	Gap
200	2	40	2.534	2.386	2.016	0.370
200	8	10	2.317	1.938	1.858	0.081
800	8	2.5	1.956	1.676	1.676	0.000
3200	8	0.6	1.755	2.533	1.603	0.930

Table 3: Width 32, hints on. Ratio is corpus size divided by tokens trained. The last row is past one pass over the data.

4.5 Each doubling of the corpus

With random 256-character pieces of the corpus, the gains per doubling are 0.040, 0.052, 0.050, 0.060, 0.099, 0.062 and 0.061 for leave-one-out with dropout, and 0.053, 0.055, 0.052, 0.060, 0.073, 0.028 and 0.018 for earlier steps (Figure 1). Neither curve shows gains that rise with each doubling, so we did not reproduce the blog post's result on Shakespeare. In the earlier-steps model the loss on

hinted characters stops falling after 251K characters, about 2.5 times its tokens trained, because it never learned how much to trust larger counts. In the leave-one-out model it keeps falling to 1.66.

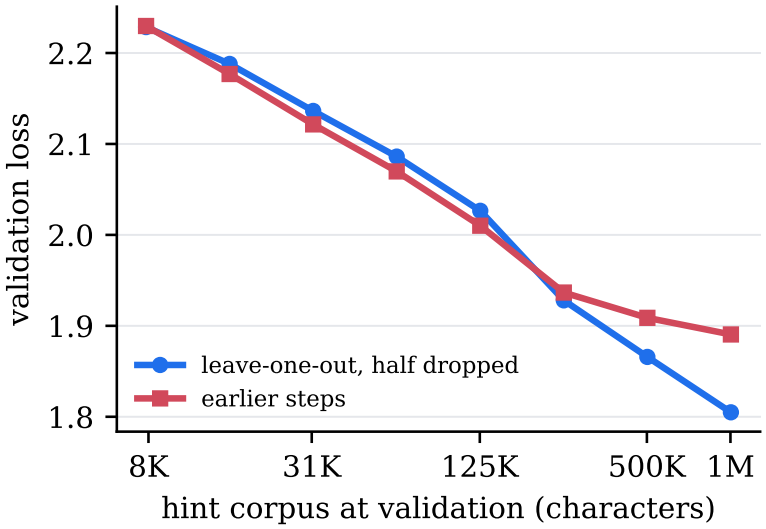


Figure 1: One final model of each kind evaluated with random subsets of the corpus. Width 64, 200 steps.

5 Web Text

We repeated the test on FineWeb [1], using GPT-2 tokens from one 100M-token training shard as the corpus and text from the separate validation shard, with corpus sizes from 0.78M to 100M tokens. The model has width 40, 4 layers, about 1.4M parameters, batch 8 and 90 steps (46K trained tokens) unless stated. Hints use lengths 6, 10 and 20 tokens unless stated; the longest set adds 2, 3, 4 and 5.

5.1 The hints improve with each doubling on web text

Before any model is involved, the share of validation tokens where the hint is right grows faster with each doubling: each doubling adds 0.07% of tokens at 1.56M and 0.71% at 100M, where the hint is right on 2.06% of tokens. This matches the blog post's report for the hints, unlike our Shakespeare result. We did not check the cause; near-duplicate pages in both shards are one possibility.

5.2 Leaving out the whole page removes the harm on small corpora

An earlier sweep with 256 validation chunks compared hints from earlier steps with leave-one-out and half dropped: 7.860 against 7.835 at 100M, but leave-one-out lost below 25M (7.884 against 7.889 at 0.78M), and the 46K-token stream found 0.36% of tokens with a hint or fewer, so earlier-steps hints barely trained. Validation loss without hints was 7.871. Removing only a token's own occurrence does harm below 25M tokens: the overall loss is 7.877 at 0.78M. Removing every match from the token's own page removes this harm. At 100M the loss is 7.816 with the page left out and no hints dropped, against 7.821 with one occurrence left out and half of the hints dropped, and with the page left out dropping half no longer helps (7.825).

5.3 Short matches give most of the gain, and that gain shrinks with training

Adding matches of 2, 3, 4 and 5 tokens raises the share of tokens with a hint from 5% to 92% and gives by far the largest drop, from 7.816 to 7.244 at 100M. Most of this comes from lengths 2 and 3: length 2 alone gives 7.373, and length 4 and longer gives 7.696.

Table 4 shows longer training at corpus 100M. The gain from short matches falls from 0.63 to 0.34 over 16 times more training, because the model learns common next tokens by itself. The gain from lengths 6, 10 and 20 stays between 0.04 and 0.07. The model with all seven lengths depends on them: with hints off it scores 7.754 against 6.916 at the longest run. So this gain is recall during validation and not faster learning, since the model's own predictions are worse than those of a model trained with no hints.

Steps	Tokens trained	No hints	Lengths 6, 10, 20	All seven lengths
90	46K	7.871	7.816	7.244
360	184K	7.398	7.333	6.879
1440	737K	6.916	6.872	6.580

Table 4: Validation loss at corpus 100M. Hints leave out the token's own page. The vocabulary grows with the steps (28,359, 39,394 and 46,243 tokens), so compare within a row.

6 Facts and Reasoning

Engram [7] reports gains on reasoning benchmarks as well as knowledge benchmarks, with learned tables, and RETRO's gains largely come from tokens shared with the test data [22]. We found no test of retrieved next-token hints by task kind, so we built a language where the kind of each answer is known.

Setup. The corpus holds 1.5M lines of made-up text: rare facts stated 3 times each, common facts stated 300 times, sums modulo 10 with 20 of 100 pairs held out, two-step comparisons with fresh names, and reading an answer from the line. Hints use lengths 2, 3, 4 and 6, leave out each token's own occurrence, and enter after the embedding, after layer 2 and before the output head. The model has width 64 and 4 layers, and trains for 5760 steps of 32 lines at learning rate 0.02. We test 1000 new lines per group. Chance is 1% for facts, about 10% for sums and 50% for reading and comparison, where always picking the first name scores that much.

Hints	Common	Rare	Other words	Not in table	Held-out sum	Read	Comparison
None	.995	.009	.005	.013	.492	.480	1.000
All 3 places	1.000	1.000	.010	.004	0	.502	.495
All 3 places, half dropped	1.000	1.000	.006	.012	0	.513	.495
Input only	1.000	1.000	.006	.012	0	.513	.497
Length 6 only, all 3 places	1.000	1.000	.012	.010	.730	.474	.499
Output only	1.000	1.000	.010	.003	0	.924	1.000
Output only, half dropped	**1.000**	**1.000**	**.006**	**.012**	**.536**	**.499**	**.999**

Table 5: Accuracy at the answer token.

Results. Table 5 shows that hints take rare facts from chance to 100% when the test uses the stored words, and give nothing when the words change or the fact is absent. Hints at all three places stop the model from learning comparison (.495 against 1.000) and held-out sums (0 against .492). Hint dropout alone does not fix this. Hints only at the output restore comparison, and with half dropped they also restore sums and keep all recall. Section 8 tests why.

This does not contradict Engram: its tables are learned, while our hints are retrieved next tokens. A made-up language shows a mechanism, not its size in real models.

6.1 Web text and reasoning in one model

Next we trained one model on both languages at once, with 2 web chunks and 32 made-up lines per step for 5760 steps. The two languages share no tokens, so one joined hint table gives the same hints as two tables. Web hints use all seven lengths and leave out the token's page, with corpus 100M.

Width	Hints	Web, hints on	Web, hints off	Held-out sum	Comparison
32	None	–	7.132	.493	.600
32	All 3 places	6.824	7.846	0	.496
32	Output only, half dropped	7.095	7.107	.458	.929
64	None	–	6.991	.350	.971
64	All 3 places	6.714	7.637	.048	.496
64	Output only	6.878	7.061	0	1.000
64	Output only, half dropped	6.971	6.991	.267	.995
96	None	–	6.956	.484	.981
96	All 3 places	6.677	7.602	0	.495
96	Output only	6.777	6.988	0	.995
96	Output only, half dropped	6.863	6.942	.434	.996

Table 6: One model trained on web text and the made-up language. Accuracy at the answer token. Rare facts in the table scored 1.000 in every hint row.

Table 6 shows a trade-off. Hints at all three places give the largest web text gain, 0.28 to 0.31, and block comparison at all three sizes; with hints off their web loss is 0.65 to 0.71 worse than no hints. Output-only hints keep comparison and keep 41% of that gain at width 64 and 64% at width 96. Dropping half restores held-out sums too but keeps only 12%, 7% and 33% of the gain at widths 32, 64 and 96.

7 How Rare Text Must Be for Hints to Help

Recall in language models tracks how often a fact appears [15, 16], and one study reports verbatim recall near 64 exposures at 1B parameters and 16 at 8B [17], while another finds the effect disappears on some models [18]. For retrieval, kNN-LM papers disagree on whether rare patterns [19] or high-frequency tokens [20] gain most. We found no such split for retrieved next-token hints.

7.1 A model needs only a few statements of a fact

Each of 200 facts per group was stated exactly 1, 2, 4 up to 256 times in training, with nothing else in the data. The table holds every fact, so a hint is always right. Table 7 shows accuracy without

hints. The model reaches 50% at about 45, 19, 10, 5, 4, 3, 3 and 2 statements at widths 16, 24, 32, 48, 64, 96, 128 and 192, so a wider model needs fewer statements. This agrees in direction with [17] and with a width bound for memorizing facts [27]. Going from 2 to 8 layers did not move it, so the count does not follow the number of weights. With 4 times more training the same count gives the same accuracy, so the count matters and not the length of the run. With hints every group scores 100%, but a model trained with hints scores at most 2.5% with hints off, even at 256 statements: it memorized almost nothing itself.

Statements in training	1	2	4	8	16	32
Width 16	.000	.035	.005	.065	.105	.260
Width 24	.020	.025	.075	.175	.395	.885
Width 32	.030	.045	.130	.380	.820	.995
Width 48	.055	.130	.350	.775	1.000	1.000
Width 64	.060	.165	.535	.870	.990	1.000
Width 96	.120	.275	.670	.935	1.000	1.000
Width 128	.125	.310	.735	.970	1.000	1.000
Width 192	.185	.435	.775	.965	.995	1.000
Width 64, 2 layers	.070	.175	.470	.850	.995	1.000
Width 64, 8 layers	.075	.175	.530	.865	1.000	1.000
Width 64, 400 facts per count	.068	.145	.360	.770	.965	1.000
Width 64, 4 times more training	–	–	.485	.885	.995	1.000

Table 7: Accuracy without hints by how many times a fact was stated. Chance is 1%.

7.2 Keeping a model memorizing while it uses hints

A model that always gets hints stops memorizing, so we tried ways to keep both, at width 64 (Table 8). Taking the hints away from a random share of tokens works only at a high share: 25% still leaves 70% accuracy at 256 statements, and 90% matches the model that never saw hints. Two methods keep 100% with hints and full memorizing. Training every step on the same batch with and without hints matches the model that never saw hints at every count, for 28% more clock time. Keeping only text seen under 16 times in the table gives 100% without hints at 16 statements and above, where the model gets no hint. The closest work we found drops privileged inputs in reinforcement learning [28], and R-Drop also runs two passes per batch, on the same input [29].

7.3 Only training with and without hints survives web text and reasoning

We repeated the best methods in the joint model at width 64 (Table 9). Training every step with and without hints keeps 96% of the web text gain with hints on, and with hints off it beats the model that never saw hints by 0.039 and recovers sums and comparison, for 47% more clock time. With hints on, reasoning stays blocked, and a common fact reaches only 7% with hints off against 78% without hints. A table of only rare text loses the web text gain and comparison stays at chance.

7.4 On web text the gain sits on text the model itself has not seen

We split the gain of 0.357 at 1440 steps over 65K validation tokens (Table 10). Gain per token rises with match length, from 0.22 at length 2 to 3.98 at length 20, but lengths 2 to 4 hold 72% of the gain because they cover far more tokens. Hints that are right, on 18% of tokens, give 125% of the gain, and wrong hints take back 20%.

Statements in training	1	4	8	16	32	64	256
No hints in training	.060	.535	.870	.990	1.000	1.000	1.000
Hints always	.000	.000	.000	.000	.000	.005	.005
25% of tokens without hints	.005	.010	.015	.005	.010	.140	.700
50% of tokens without hints	.020	.010	.030	.060	.285	.915	1.000
75% of tokens without hints	.045	.235	.555	.960	1.000	1.000	1.000
90% of tokens without hints	.070	.425	.785	.985	1.000	1.000	1.000
No hints in last half	.045	.225	.515	.865	.995	1.000	1.000
No hints in last quarter	.005	.040	.055	.175	.375	.765	1.000
Hints at output only	.020	.000	.010	.000	.010	.030	.010
Every step with and without hints	.065	.475	.890	.985	1.000	1.000	1.000
Table keeps text seen under 16 times	.015	.025	.000	1.000	1.000	1.000	1.000

Table 8: Accuracy with hints off, width 64, one run each. With hints on, every row scores at least 99% except no hints in the last quarter.

Training	Web on	Web off	Sum on/off	Comparison on/off	Fact off	Gain
No hints	–	6.991	.350	.971	.780	–
All three places	6.714	7.637	.048/.000	.496/.014	.012	100%
With and without hints	**6.726**	**6.952**	.040/.403	.500/.960	.070	96%
Output only, with and without	6.924	6.980	.270/.270	.979/.979	.036	24%
75% of tokens without hints	6.868	6.991	.222/.222	.914/.774	.270	44%
Table under 16 times	7.010	7.077	.581/.581	.495/.505	1.000	0%

Table 9: One model on web text and the made-up language, width 64, 5760 steps, one run each. Hints enter at all three places unless named. Gain is the share of the 0.277 web text gain of hints at all three places. Fact off is a common fact with hints off.

The model trained on only 737K tokens, 1/135 of the table, so text seen 16 times in the table is usually text the model never saw. Table 11 counts what the model itself saw: 79% of the gain sits on contexts it never saw, and gain per token falls from 0.42 at zero sightings to about zero at 64 or more, which agrees with the facts test. These models sit at a loss near 6.9, so this describes early training only.

8 How Hints Block Reasoning

8.1 Training with hints from the start blocks the skill more than adding hints late

To find the cause we built a test that runs in seconds: 2 layers and 1000 steps on half facts and half comparison lines "a > b . b > c . top ? a .", with the two statements in random order and name triples held out of training. Hints go in at all three places. At width 32 the shared table gives comparison .545, and the model says the first name in the line 94% of the time, which is right half the time.

We trained a model with no hints for 600 steps, then trained only the hint trust for 200 steps with the model frozen, then trained everything with hints on for 800 more steps (Table 12). Hints at full strength took facts from about .06 to .80. Comparison stayed level at width 32 and at 4 layers (.946 to .958, .931 to .950) and fell at widths 64 and 128 (.974 to .853, .996 to .809). Further training with hints then removed the skill at width 32 and at 4 layers (.504, .580) and left it about

Group	Share of tokens	Share of gain	Gain per token
No match	8%	−4%	−0.17
Longest match 2 to 4	80%	72%	0.32
Longest match 5	7%	11%	0.56
Longest match 6	4%	12%	1.05
Longest match 10	0.6%	5%	3.16
Longest match 20	0.3%	3%	3.98
Match seen once	19%	28%	0.53
Match seen 2 to 15 times	35%	39%	0.40
Match seen 16 or more times	38%	37%	0.35

Table 10: Web text gain of hints of all seven lengths. The two blocks split the same tokens.

Times the model saw the context	Share of tokens	Share of gain	Gain per token
0	67.2%	79.2%	0.421
1	7.9%	8.3%	0.377
2 to 3	4.7%	5.1%	0.393
4 to 15	5.0%	3.7%	0.265
16 to 63	2.3%	0.9%	0.138
64 or more	1.5%	−0.1%	−0.026

Table 11: Web text gain split by sightings in the model's own 737K training tokens, for the 88.6% of tokens whose longest match fits inside their chunk.

the same at widths 64 and 128 (.852, .872). A model trained with hints from the start was lowest or level with the lowest at every size (.505, .691, .522, .499). So the block comes more from training with hints than from the hints alone, and adding hints late saves the skill only at the two larger widths. Hints with the shared table already solve training lines: after 600 steps of always-on hints the model scores .999 on them and .52 on held-out names.

Hand settings of hints point the same way but do not settle the cause. In runs trained with and without hints, comparison is .764 with hints everywhere, .996 with hints only at the answer word and .516 with hints everywhere else. Turning off one hint, at the dot between the statements, gives 1.000 at width 32 but .553 at width 128.

Comparison	Width 32	Width 64	Width 128	4 layers
Trained with no hints, hints off	.946	.974	.996	.931
Then hints on, only trust trained	.958	.853	.809	.950
Then 800 more steps with hints on	.504	.852	.872	.580
Hints on from the start	.505	.691	.522	.499

Table 12: Comparison on held-out names with hints on unless stated, one run each.

8.2 A small table in training keeps reasoning and improves facts

Brunborg trains with a small corpus and validates with a large one, because a large corpus from the start makes the model rely on hints too much [1]. We tested that choice on reasoning (Table 13). The training table reads a share of the lines the model does not train on, the test table reads all of them, and test counts are scaled to the same count levels so that trust learned in training applies.

Hints stay on, with no switch and no second pass. The model trains on 16,000 comparison lines and the full table holds 84,000 others, so 5% to 60% is 0.25 to 3 times the lines trained on. In this range comparison is .97 or above at all three widths and facts reach about .88, above the shared table's .81. At 2% comparison falls at width 32. From 70% to 95% the result is erratic, and at the largest table comparison is back at chance at every width. We cannot explain the break. Training lines score near 1.0 in the good and the bad runs, and from 20% up the hint at the answer word is right equally often on training lines at every size (.50 for the longest hint, .55 for any length), so the hint at the answer does not explain it.

A separate table also works. With the table reading one half of the name triples and training using the other half, comparison is 1.000 at widths 32, 64, 128 and at 4 layers, and at 4000 steps at width 32, with facts at .80 to .82. With hints off comparison stays low (0 to .23) in these runs, so the model needs its hints.

Share of table	Times the lines trained on	Width 32	Width 64	Width 128	Facts
2%	0.1	.781	–	.957	.81–.87
5%	0.25	**.996**	**.996**	**.994**	.88–.89
10%	0.5	**1.000**	–	**1.000**	.88–.89
20%	1	**1.000**	**1.000**	**1.000**	.87–.88
30%	1.6	**1.000**	**1.000**	**1.000**	.87
40%	2.1	**.997**	**.990**	**.990**	.87
50%	2.6	**.999**	–	**1.000**	.88
60%	3.2	**.972**	**.984**	**1.000**	.89
70%	3.7	.897	.945	.785	.89
75%	3.9	.549	–	1.000	.89
80%	4.2	1.000	.877	1.000	.88–.89
90%	4.7	.986	.976	.823	.88–.89
95%	5.0	.982	.504	.541	.89
100%	5.3	.523	.597	.503	.89

Table 13: Comparison with hints always on, 1000 steps, one run each, all testing with the full table. Bold marks the range where every width is at .97 or above. We did not run this on web text or on sums.

8.3 The limit follows copies in the table, not table size

The break could depend on table size against the data trained on, or on how many copies of each line the table holds. We kept the same 84,000 table lines and 16,000 training lines and changed the number of names, which changes copies of each line and nothing else (Table 14). With 30 or 40 names the full table no longer blocks comparison, and facts rise to .97 and .99. Comparison stays high at few copies and falls at many; the next subsection maps the limit. Changing training steps fits copies better than the ratio: 2000 steps is 2.1 times the lines trained on, inside the safe ratio, but 5.8 copies, and gives the erratic .920 and .720.

Several explanations failed. Length 6 hints are the only ones that hold all three names, but without them the full table still blocks comparison (.504, .508 and .448 at widths 32, 64 and 128), and length 4 hints alone block it too (.492 and .501). Giving every hint one trust level, with no count or share signal, leaves the 100% table blocked (.590, .529) and the 20% table working (.982, .978), with facts at .89, so the count scaling above is not needed here. A 20% table used in training and testing keeps comparison at 1.000 with facts at .66 and .75, so the block comes from the table in training and the full table at test adds only facts. In a blocked model, removing the hint at the

word "top" raises held-out comparison from .52 to .80 at width 32 and from .50 to .61 at width 128. We map the limit and look for the cause next.

Setting	Copies of each line	Width 32	Width 128
14 names, full table	35	.552	.560
20 names, 500 steps	7.9	.552	.884
20 names, full table	7.2	.523	.503
20 names, 90% of table	6.5	.986	.823
20 names, 2000 steps	5.8	.920	.720
20 names, 60% of table	4.3	**.972**	**1.000**
30 names, full table	1.8	**.999**	**1.000**
40 names, full table	0.7	**1.000**	**1.000**

Table 14: Comparison with hints always on, one run each. The 14, 20, 30 and 40 name full tables hold the same number of lines. A copy here is one more table line with the same three names in the same order and the same statement order, so each triple has two versions.

8.4 Copies of the training sentences block the skill at every name count

Table 15 sets copies directly at four name counts. Up to 5 copies comparison was .94 or above in all 32 runs, 7 and 10 copies gave mixed scores, and at 15 or more copies 11 of 14 runs stayed near .50. Copies of the same sentence fit better than table size: the 40 name table at 15 copies holds 1.75 million comparison lines and the 14 name table at 30 copies holds 71,000 lines, and both block. These runs scale test counts by table lines over all lines (`-countscale 1`). Without scaling, the comparison score at 14 names falls to between .51 and .72 for 1 to 7 copies, and facts are .73 and .89 at 14 and 20 names, against about .1 to .2 at few copies with scaling, so the low copy cells at 14 names depend on how counts are matched. The limit of 5 copies therefore holds only with test counts scaled. Scaling has its own cost: in 12 of the 32 runs at 5 copies or fewer (every 14 name run, and 20 names at 1 and 2 copies) facts with hints were only .08 to .22, so those models mostly ignored their hints and those cells do not show hints and comparison working together. In the other 20 runs facts were .88 or above.

Names	1	2	4	5	7	10	15	30
14	.95/.95	.99/.99	1.0/1.0	.99/.99	.96/.98	.56/.53	.98/.50	.54/.57
20	1.0/1.0	1.0/1.0	1.0/1.0	1.0/1.0	.81/.98	.51/.71	.69/.52	.65/.48
30	1.0/1.0	1.0/1.0	1.0/.99	.99/.99	.99/.93	1.0/.54	.51/.49	.48/.51
40	1.0/1.0	1.0/1.0	1.0/1.0	1.0/1.0	.99/1.0	.84/.49	.48/.48	–

Table 15: Comparison score at width 32 / width 128 by copies of each line in the table, with hints always on, one run each. A dash marks runs we skipped because the table is too large.

Only copies of the sentences being trained on matter. When the table reads one half of the name triples and training uses the other half, comparison is 1.000 at both widths at 17 and at 60 copies of each table line, with facts at .80. Repeating training lines does not block either: 1000 lines trained 16 times each, with 2 copies in the table, score .999 and 1.000. Blocked models also read the table: with 30 copies of the training lines, adding the held-out test sentences to the test table 30 times raises their score from .65 and .48 to 1.000 at both widths, while adding them 7 times gives .67 and .48. So the blocked model has not learned comparison. It depends on the hints a sentence gets when the table holds it about as often as the training sentences.

The blocked model uses a simple shortcut. On held-out lines it says the first name in the line, which is right when the first statement comes first and wrong otherwise, so it scored 1.00 and about .00 on the two kinds of line, and .50 overall. On its training lines it scores 1.00 on both.

Removing hints at single word positions on comparison lines, in training and testing, shows where the block starts (20 names, 30 copies, widths 32 and 128). Without the hint at the first dot, the word between the two statements, comparison rises from .65 and .48 to .95 and 1.000, and across 14, 30 and 40 names it stayed above .90 in 5 of 6 runs. Removing the hint at any of the other single words left at least one width blocked; removing the word 7 hint gave .92 and .48. With only the answer hint kept, comparison reaches .996 and .999, so the answer hint alone does not block. Removing the answer hint does not fix the block (.48 and .65), so the block does not come from copying the answer hint. We do not know why the hint at the first dot matters.

The block held as models grew. At 30 names, 2 copies gave 1.000 at widths 32, 64, 128 and 256, at 4 layers and at 4000 steps. At 30 copies comparison stayed at .48 to .51 at those four widths and at .50 at 4000 steps at width 128, but it escaped twice: .995 at 4 layers and .973 at 4000 steps at width 32. Each result is one run, on a made-up language.

8.5 A switch learns to shut hints only when taught

We multiply each token's hint by a switch between 0 and 1, read by a linear layer from the hidden state of a pass without hints. Learned from the loss alone, the switch stays at 1.0 and comparison stays at .769. We then gave it a second target: predict whether the longest hint equals the next token. This target is known from the data and adds one small loss. In the small test the taught switch gives comparison between .50 and .996, and no place in the model does best at every size.

8.6 The taught switch also lowers web text loss

We added the taught switch to the joint model, at all three places, training every step with and without hints (Table 16). Comparison with hints on rises from chance to .905, 1.000 and 1.000. Web text loss with hints falls below plain hints by 0.043, 0.070 and 0.094 at widths 32, 64 and 96, so the switch does not trade the web text gain for reasoning. At width 64 this is 125% of the plain gain, and training with and without hints alone reaches 96%, so the extra gain comes from the switch. A common fact with hints off reaches .621 and .957 at widths 64 and 96 against .012 and .011 with plain hints. The switch averages .24 to .27 on web text hints. Held-out sums with hints on stay blocked (0, .251, .094). With the switch at the input only, width 64 gives 6.671 on web text but comparison stays at .495, the reverse of the small test.

The lead depends on training length. At width 64 the switch is behind at 360 steps (7.155 against 6.921), level at 1440 (6.819 against 6.816) and ahead at 5760 (6.644 against 6.714), and comparison is not yet learned below 5760 steps. At 1440 steps a switch with no training pass without hints gives no gain and comparison .007; we did not run that arm at 5760 steps. The widths span only 3.2 times the parameters, each setting ran once, and the switch needs a pass without hints at every step, so it costs clock time we did not measure under equal load.

The closest work we found has no such target. Engram's gate reads the hidden state and the memory [7] and we found no separate target for it in the paper. kNN-Adapter learns a per-token mix on top of a frozen pretrained model [24], and SpecDec++ trains an acceptance head for speculative decoding at inference [26]. We found no gate trained during pretraining on whether the retrieved token is right.

Width	Web, no hints	Web, plain hints	Web, switch	Web, switch off	Comparison, plain/switch	Fact off
32	7.132	6.824	6.781	7.082	.496/.905	.014
64	6.991	6.714	**6.644**	6.952	.496/1.000	.621
96	6.956	6.677	**6.582**	6.918	.495/1.000	.957

Table 16: One model on web text and the made-up language, 5760 steps, one run each. Fact off is a common fact with hints off in the switch model.

9 Hints on a Finished Model

All earlier sections train the model with hints. Here we leave a finished model unchanged and only raise the output score of each hinted token, so training cannot remove a skill. The model is MiniCPM5-2B (2.5 billion parameters, 42 layers, Apache 2.0), in its 4-bit MLX form on a laptop. The closest earlier work is kNN-Adapter [24], which mixes a finished model's output with retrieval from a target-domain store.

The table stores the most common next token after word groups of 2, 3, 4 and 6 tokens. A learned trust number sets how far each hint raises the score. It depends on the hint length, how often the group was seen, how often that token followed it, and how likely the model already thinks that token is. Trust has 840 numbers and trains in about a second from about 10,000 places. Except for the row that tests pages stored in the table, table, trust and test text never share a file or page. The model reads 128 tokens at a time and each setting ran once on about 8,000 test places unless stated. All code is in `finished/`.

9.1 Hints recall stored text but miss reworded questions

The table read 70% of about 52,000 Wikipedia pages from an April 2026 copy [25], plus 48 pages about 2026 events; 27.6M tokens and 24.8M groups of 6 tokens. Trust trained on 15% of the pages and the test used the rest (18,840 places).

Test	Hints off	Hints on
Unseen Wikipedia text, next word right	41.7%	43.1%
Unseen Wikipedia text, loss	3.145	3.086
2026 pages stored in the table, next word right	41.6%	83.0%
11 new facts, the page's own words	1 right	6 right
The same 11 facts, reworded	1 right	1 right
100 comparison questions (three names)	54 right	54 right
100 two-digit sums	1 right	1 right

Table 17: MiniCPM5-2B with and without hints. No comparison answer changed. The 4-bit model given a bare sum fails with or without hints, so the sums say nothing about harm.

The row for 2026 pages tests lookup, not prediction: those pages are in the table, so most long matches return the page's own next word. The 11 facts were written by hand from pages in the table, once in the page's own words and once in other words. A reworded question misses the stored word group or meets a wrong hint: "the operating system on the Acer Googlebook 14 is" went from "Windows 10" to "a Linux-based", and the page says Google. For new facts asked in new words, search over documents remains the better tool.

9.2 The gain rises with exact repeats and with how low the model scores alone

We repeated the test on other text, with a new table and trust each time. Table 18 is sorted by the share of places where a 6-token hint exists. The share does not set the order of the gains by itself: pandas at 38% gains 1.9 points and the notes at 20% gain 11.

Text, table and test from different files	Table tokens	Off	On	6-token hint
transformers code, unseen model folders	6.3M	66.2%	**83.9%**	81%
transformers code	5.3M	65.7%	**79.2%**	69%
pandas code	3.8M	67.0%	68.9%	38%
sympy code	6.0M	64.4%	67.9%	35%
sklearn code	2.4M	64.8%	68.0%	34%
networkx code	1.3M	65.2%	68.7%	33%
scipy code	4.3M	61.3%	63.0%	24%
Own Markdown notes, exact copies removed	5.6M	32.1%	**43.1%**	20%
sympy code, 10% of table	0.45M	64.4%	65.1%	13%
Wikipedia, full table	27.6M	41.7%	43.1%	10%
Wikipedia, 10% of table	2.6M	42.0%	42.6%	4%
Table from transformers, test on torch	7.3M	58.6%	58.4%	4%
sympy code, 1% of table	0.04M	64.4%	64.4%	3%
Wikipedia, 1% of table	0.26M	42.0%	42.1%	1%

Table 18: Next word right with hints off and on, by kind of text.

Three things follow. A table from a different source gives nothing: transformers code did not help torch code. Holding out whole model folders did not lower the transformers gain, likely because model files in that library copy each other's code; transformers is the extreme case, and five other libraries gain 1.7 to 3.5 points. The notes gain 11 points at only 20% because the model starts low there, at 32.1%, on private text it has never read. With exact duplicate files left in, 27.6% of the notes by size, that row read 33.7% to 56.9%, so we removed them; near copies remain.

The share of 6-token matches can be measured from a sample of text with no model at all. In these runs, 24% or less on code and Wikipedia gave under 2 points, and so did pandas at 38%. The share is a cheap first filter and not a rule, since the notes at 20% gained 11 points and sympy at 35% gains little once the model reads more (next subsection). Small tables raised the loss slightly even with trust trained on the same kind of text (sympy 1.708 to 1.725 at 10% of the table and 1.755 at 1%; Wikipedia 3.126 to 3.133 and 3.144), and so did the table from another library (2.093 to 2.136). A table that raises the loss should be left off.

On Wikipedia the gain sits where several hint lengths agree. Where all four fire, 10% of places, the top word rises from 48.5% to 57.2%; where three fire, 41.9% to 43.5%; with one or two the change is 0.5 points or less. It also sits where the model is unsure: 13.5% to 15.8% when its top word holds 10% to 25% of its probability, and 91.5% to 91.6% when that word holds over 90%.

9.3 More context removes most of the gain on ordinary code

A real assistant reads more than 128 tokens, so we repeated two code rows with the model reading 512 tokens and scored the last 120 places of each block (5,000 to 8,000 places). In sympy the model alone rose from 64.4% to 79.1%, and hints added only 0.8 points to reach 79.9%, with loss unchanged (0.904 and 0.907). In transformers unseen folders the model rose from 66.2% to 77.1%, and hints still added 8.6 points to reach 85.7% (loss 0.993 to 0.646). So on ordinary code, reading more text gives far more than a table does; only heavily copied code keeps a clear gain.

9.4 Whole phrases and cost

Single next words overstate a user's gain. Asking for 5 tokens in a row, 96 prompts each, the transformers split rose from 25 to 40 right, and sympy stayed level at 26 and 25.

A lookup took 0.66 ms per token in Python on the laptop, against 5.6 ms per token for the model reading 128 tokens at once. A table of 6M tokens took 234 MB of arrays and about 5 s to build, so a repeated text source can be added or rebuilt without touching the model.

9.5 A finished model gains little on general text because exact matches are rare

The model does not already know this text: its top word is right only 41.7% of the time on unseen Wikipedia. The gain is small because long exact matches are rare. Where all four hint lengths fire, 10% of places, the top word gains 8.7 points, which is most of the 1.4 point total. Over training, the loss gain also shrinks as a model learns. The width-40 model gained 0.63 loss at 90 steps, 0.52 at 360 and 0.34 at 1,440, and contexts it had never seen gained 0.42 per token against -0.03 for contexts seen 64 or more times (Table 11). The finished model gains 0.06 on Wikipedia. The comparison is rough: the text, table size and hint place all differ. Hints lower the loss most early in training, and at inference for repeated text that the model has not learned. The early drop is recall and not learning, because those models score worse than models trained with no hints when the hints are off.

10 Limits

- The web text models stop at a loss far above real models, so they do not show what hints do for a trained model. The largest joint model has 4.3M parameters.
- Each setting was run once, and learning rates were picked on the same validation chunks used for reporting.
- Reasoning in this paper means one comparison task in a made-up language, plus held-out sums. It does not stand for reasoning in general.
- We did not run hints from earlier steps on the comparison task, so we do not know whether they keep it.
- The speedrun record measures time to a fixed early loss. It says nothing about final quality, and neither do our runs at equal steps.
- On the finished model, near copies between files remain in the table and test, the fact test has only 11 facts, and the model is 4-bit.
- Shakespeare is not web text: it has no crawled duplicates and its tokens are characters, so our result there does not contradict the blog post's rising returns.
- In a real run, leave-one-out needs the full corpus before training starts, or a table large enough to subtract one count per lookup.
- We measured loss at equal steps, not clock time, except where stated.

11 Open Questions

1. If a finished large model reads code with a table of that code, does a user accept more completions than with search over the same code?

2. If hints match similar word groups instead of exact ones, do reworded questions recall new facts on a finished model?

3. If a GPT-2-sized model is trained with Muon on web text, does adding n-gram hints from the training data lower validation loss compared with the same model without hints?

4. If a model is trained on web text with a hint corpus about 500 times larger than the tokens it trains on, does leaving out the token's page and dropping half of the hints beat hints from earlier steps?

5. If a model is trained on web text with hints, does each doubling of the hint corpus lower validation loss by more than the doubling before it, after near-duplicate documents are removed?

6. If a model is trained with hints, does it repeat training text word for word more often than the same model without hints?

7. If a model trains far longer than our web text runs, does the validation loss gain from short matches disappear?

8. If a hint table drops n-grams that the model itself has already seen often, how much of the web text gain is kept?

9. If a GPT-2-sized model is trained on web text with a switch taught to predict whether each hint is right, is its validation loss lower than with plain hints, and does the lead keep growing with training length?

10. If the switch is trained with no pass without hints, does it still lower web text loss at long training?

11. If a hint table is capped so that it holds at most a few copies of each passage, does a model trained on web text keep reasoning tasks and the web text gain?

12. If a hint table holds passages the model trains on 15 or more times, does a model trained on web text lose reasoning skills that it keeps with 5 or fewer? Why does the hint at the word between two statements start the block in the toy?

13. If a model is trained on web text and reasoning tasks with 10% to 50% of the hint table and tested with all of it, does it keep comparison and sums, and how much of the web text gain does it keep?

12 Conclusion

For hints from a full corpus, remove each token's own occurrence, and on web text remove its whole page. On Shakespeare, dropping half of the hints as well beat hints from earlier steps by about 0.08 at three sizes, and by more when the corpus was far larger than the tokens trained. On web text with the page left out, dropping hints did not help (7.825 against 7.816). If a run makes more than one pass, causal hints need the same removal. These steps fix validation loss only: a full table

in training still stops the model from memorizing and from learning comparison. If hints must not replace the model's own learning, the simplest candidate is a small table in training and the full table at test time, which is the record's own choice. It kept comparison and improved facts in our small test but is untested on web text. With test counts scaled, the limit there was about 5 copies of each training sentence in the table, not table size; without scaling it did not hold at 14 names. The tested option on web text is to train every step with and without hints and add a switch trained to predict whether each hint is right. In our joint test this restored comparison with hints on and lowered web text loss further at the longest training, though sums stayed blocked with hints on. Hints at the output only need no second pass and keep reasoning, at the cost of part of the gain. These are small models and single runs, so all of this must be tested in real training on FineWeb.

On a finished model, add hints only for text that repeats word for word and that the model has not learned. The share of exact 6-token matches is a cheap first filter, not a rule. Then measure loss with and without the table on held-out text at the context length you will use, and leave off any table that raises it. On general text the largest loss drops came early in training, and they are recall and not learning, because those models score worse with hints off than models trained with no hints.

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